

Sujata.d a Notes on Aws

Amazon Web Service

Aws: AWS services is amazon's cloud web hosting platform that offers flexible, reliable, scalable, easy to use and cost effective solutions.

Cloud computing: is a term refer to storing and accessing the data over the internet

There are 3 types of cloud services:

1. Private cloud: it is dedicated to single **tenant**. It is dedicated in terms of hardware and security.
2. Public cloud: shared with multiple tenants and cost is lesser. Ex: aws, GCP (google), Oracle, Microsoft Azure, Alibaba cloud etc...
3. Hybrid cloud: hybrid is a combination of both public and private cloud. It is the most successful cloud practice. Example: Openstack & VMware

What is the reason behind cloud? Answer: Virtualization

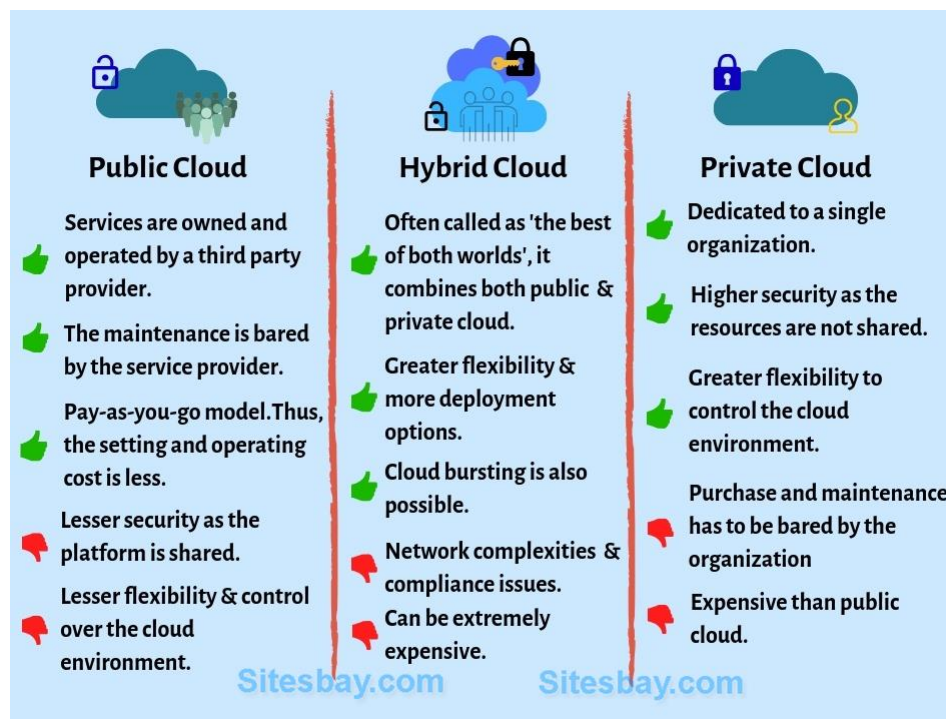
Virtualization will transfer hardware into software example: VMware

Private cloud: is having dedicated to a single organization, highly secured and Greater flexibility.

Public cloud: Third party provider makes resources and services available to the customer via Internet

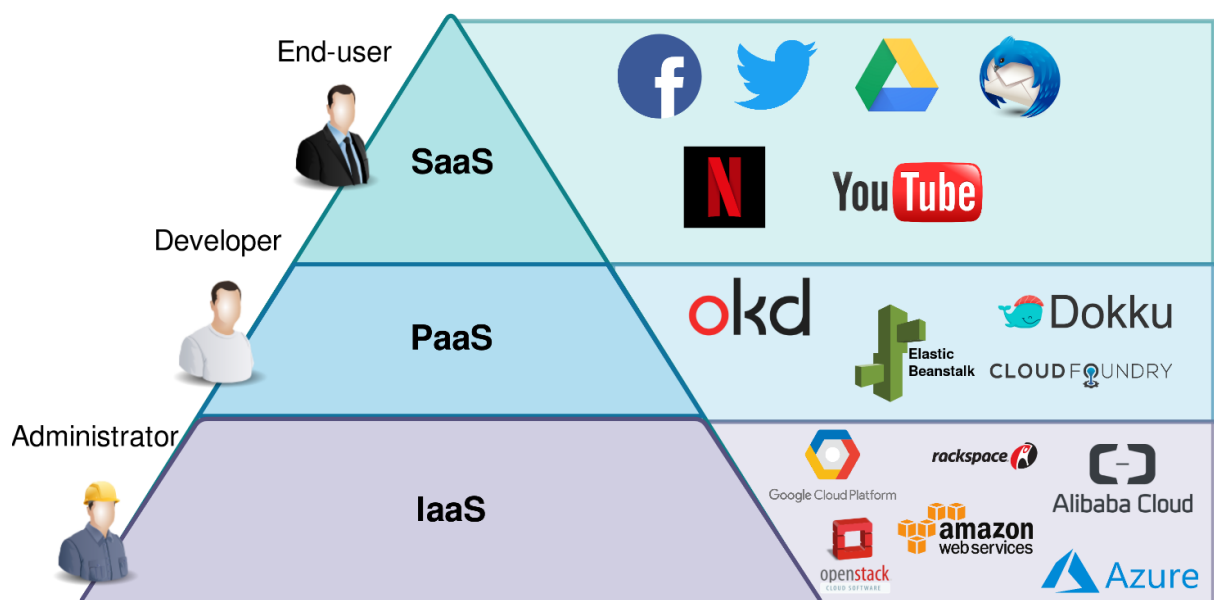
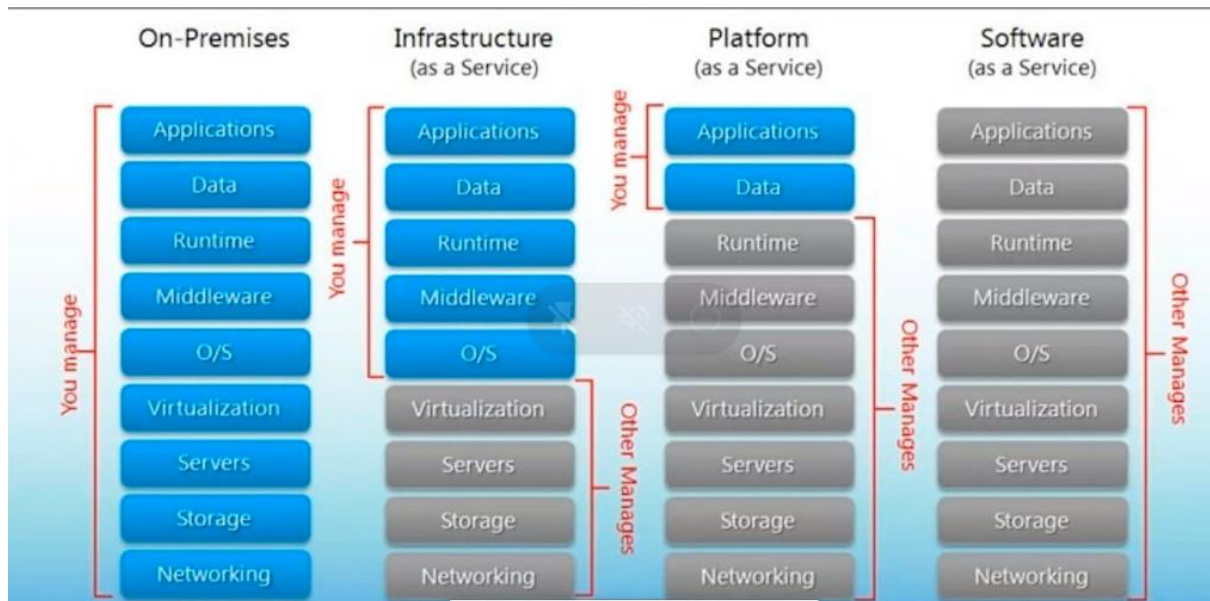
Advantages: cost effective, reliability, unlimited storage, backup & recovery

Third party providers: aws, GCP (google), Oracle, Microsoft Azure etc...



3 types of services

1. IaaS=Infra structure as a service
2. PaaS = Platform as a service
3. SaaS= Software as a service



Ip address:

Class A : 1.X.X.X TO 126.X.X.X

Class B : 128.X.X.X TO 191.X.X.X

Class C: 192.X.X.X TO 223.X.X.X

Class D: 224.X.X.X TO 239.X.X.X

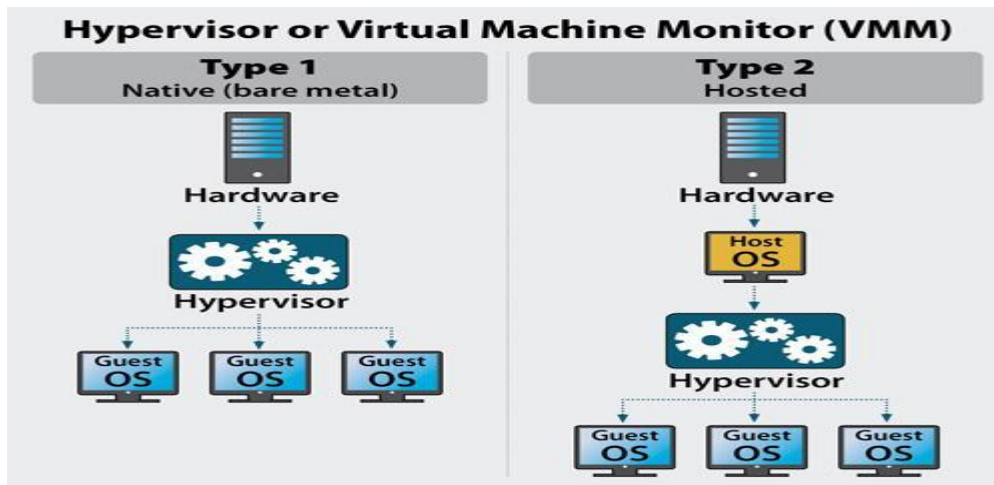
Class E: 240.X.X.X TO 254.X.X.X

127.0.0.0 is called loopback ip (it is reserved)

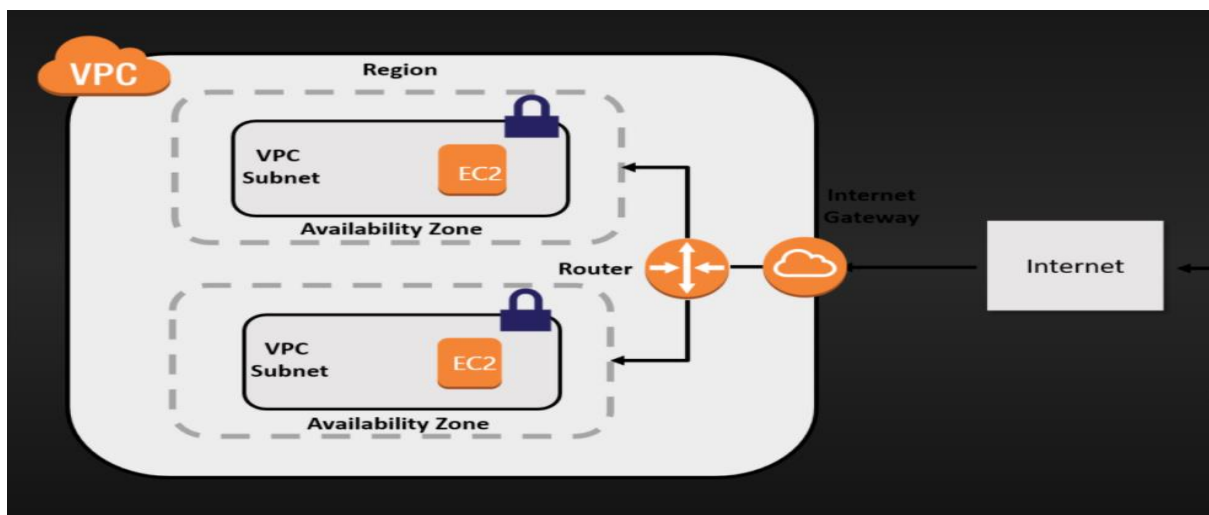
IP Header Classes:

Class	Address Range	Subnet masking	Example IP	Leading bits	Max number of networks	Application
IP Class A	1 to 126	255.0.0.0	1.1.1.1	8	128	Used for large number of hosts.
IP Class B	128 to 191	255.255.0.0	128.1.1.1	16	16384	Used for medium size network.
IP Class C	192 to 223	255.255.255.0	192.1.1.1	24	2097152	Used for local area network.
IP Class D	224 to 239	NA	NA	NA	NA	Reserve for multi-tasking.
IP Class E	240 to 254	NA	NA	NA	NA	This class is reserved for research and Development Purposes.

CIDR Notation	Host Formula	Available Hosts
/8	$2^{32-8} - 2$	16,777,214
/9	$2^{32-9} - 2$	8,388,606
/10	$2^{32-10} - 2$	4,194,302
/11	$2^{32-11} - 2$	2,097,150
/12	$2^{32-12} - 2$	1,048,574
/13	$2^{32-13} - 2$	524,286
/14	$2^{32-14} - 2$	262,142
/15	$2^{32-15} - 2$	131,070
/16	$2^{32-16} - 2$	65,534
/17	$2^{32-17} - 2$	32,766
/18	$2^{32-18} - 2$	16,382
/19	$2^{32-19} - 2$	8,190
/20	$2^{32-20} - 2$	4,094
/21	$2^{32-21} - 2$	2,046
/22	$2^{32-22} - 2$	1,022
/23	$2^{32-23} - 2$	510
/24	$2^{32-24} - 2$	254
/25	$2^{32-25} - 2$	126
/26	$2^{32-26} - 2$	62
/27	$2^{32-27} - 2$	30
/28	$2^{32-28} - 2$	14
/29	$2^{32-29} - 2$	6
/30	$2^{32-30} - 2$	2



VPC



1) What is VPC (virtual private cloud)?

It is a virtual network dedicated to your AWS account, it logically isolates from other virtual networks in the AWS cloud, and where you can launch your AWS instance.

VPC consist of subnets, network gateway & Routing table.

What is a subnet?

Subnet is a logical subdivision of IP network. The practice of dividing a network into 2 or more networks is called sub netting.

2) What is Route table?

A set of rules called routes that are used to determine where network traffic is directed.

3) **What is Internet gateway?**

A gateway that you attach to your VPC to enable communication between resources in your VPC and the internet.

4) **What is NAT gateway?**

NAT Gateway is used to connect to the Internet from instances within a private subnet in the VPC.

Nat gateway will be created from Public subnet and attach to the private subnet.

5) **What is VPC endpoint?**

It enables you to privately connect your VPC to supported AWS services and VPC endpoint services powered by private link without requiring an internet gateway, NAT device, VPN connection or AWS direct connection.

6) **What is VPC peering connection?**

A VPC peering connection is a network connection between 2 VPC'S that enables you to route traffic between them using private IPV4 or IPV6 address.

Instances in either VPC can communicate with each other as if they are within the same network.

You can create a VPC peering connection between your own VPC or with a VPC in another AWS account.

7) **Limitations of VPC peering.**

- I. You cannot create a VPC peering connection between VPC'S that have matching or overlapping IPV4 CIDR blocks. Amazon always assigns a unique IPV6 CIDR block.
- II. You have a quota on the number of active and pending VPC peering connections that you can have per VPC.
- III. VPC peering does not support transitive peering relationships. In VPC peering connection, one VPC does not have access to any other VPC with which the peer VPC may be peered.
- IV. Cannot have more than 1 VPC peering connection between the same 2 VPC at the same time.

8) **What is NACL (Network ACL)? Network Access Control Lists**

NACL are firewall at the subnet level.

You can use a network ACL to control the traffic to and from the subnet in which NAT gateway is located.

- It is a stateless means that any changes made in the inbound rule will not reflect the outbound rule, i.e., you need to add the outbound rule separately. For example, if you add an inbound rule port number 80, then you also have to explicitly add the outbound rule.

9) **What is a security group?**

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic.

When you launch an instance in a VPC, you can assign up to 5 security groups to the instance.

Security group act at instance level and not at subnet level.

- It is a stateful means that any changes made in the inbound rule will be automatically reflected in the outbound rule. For example, If you are allowing an incoming port 80, then you also have to add the outbound rule explicitly.

10) **Difference between public subnet and private subnet**

Public Subnet – Users can access resources from the internet. Internet traffic is routed via internet gateway. Applications are stored in public subnet.

Private Subnet - Users cannot access resources from the internet. Internet traffic is routed via NAT gateway. Data is stored in private subnet (database, API calls, passwords)

Types of EC2 payment models:

1. On demand or capacity reservation:
 - a. These instances work as pay as you go model.
 - b. Long time commitment is not required for this instance.
 - c. These are bit costlier
2. Spot instances:
 - a. These are called as bidding instances.
 - b. We can bid these instances as per the requirement.
3. Dedicated hosts:
 - a. Basically, dedicated instances are provided with the hardware configurations.
4. Reserved instances:
 - a. These are utilized for the longer time frame.
 - b. These would be having discounts in terms of pricing because of longer utilization time frame.
 - c. Will get to know platform, tenancy, instance type and payment options for reserved instances.

Amazon EC2 Instance Types:

General Purpose: General purpose instances provide a balance of compute, memory and networking resources, and can be used for a variety of diverse workloads. These instances are ideal for applications that use these resources in equal proportions such as web servers and code repositories.

Use cases :Developing, building, testing,

Series : T4 , T3 , t2

Compute Optimized :Instances belonging to this family are well suited for batch processing workloads, media transcoding, high performance web servers, high performance computing (HPC), scientific modeling, dedicated gaming servers and ad server engines, machine learning inference and other compute intensive applications.

Use cases : High performance computing (HPC), batch processing, ad serving, video encoding, gaming, scientific modelling, distributed analytics, and CPU-based machine learning inference.

Series : C6gn , c6gn , c5 ,c5a

Memory Optimized :Memory optimized instances are designed to deliver fast performance for workloads that process large data sets in memory.

Use cases : Memory-intensive applications such as open-source databases, in-memory caches, and real time big data analytics

Series : r6 ,r5 r5b,r5a

Accelerated Computing :Accelerated computing instances use hardware accelerators, or co-processors, to perform functions, such as floating point number calculations, graphics processing, or data pattern matching, more efficiently than is possible in software running on CPUs.

Use cases :Machine learning, high performance computing, computational fluid dynamics, computational finance, seismic analysis, speech recognition, autonomous vehicles, and drug discovery.

Series : p4 , p3 ,p2 , DL1

Storage Optimized :Storage optimized instances are designed for workloads that require high, sequential read and write access to very large data sets on local storage. They are optimized to deliver tens of thousands of low-latency, random I/O operations per second (IOPS) to applications.

Usecases : Heavy database operation

Series :Im series

AWS Storage:

Snapshot:An EBS snapshot is a point-in-time backup of your EBS volume. It is a “copy” of the data on your EBS volume. If you are looking for a disaster-recovery solution for your EBS volume, this is the solution. If you want to “backup” your EC2 instance, then you want to create EBS snapshots of the EBS volumes attached to the instance.

